

Survival Analysis with Machine Learning: A Comparative Study

Imad El Badisy^{a,*}

^a*Department of Biostatistics,
University of Medicine*

Abstract

We compare survival machine learning methods on a benchmark dataset. Cox regression serves as baseline. Results show that ensemble methods improve concordance by 5–8%.

Keywords: survival analysis, machine learning, concordance index, Cox regression

1. Introduction

Survival analysis models time-to-event outcomes, which arise frequently in clinical and epidemiological research (?). Classical methods such as the Cox proportional hazards model (?) assume a multiplicative hazard structure. Recent approaches extend this to machine learning (?).

2. Methods

2.1. Notation

Let $(T_i, \delta_i, \mathbf{x}_i)$ denote the observed time, event indicator, and covariates for subject $i = 1, \dots, n$.

2.2. Cox Proportional Hazards

The Cox model specifies:

$$h(t \mid \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}) \quad (1)$$

Parameters are estimated by maximizing the partial likelihood.

*Corresponding author

Email address: elbadisyimad@gmail.com (Imad El Badisy)

2.3. Concordance Index

Model discrimination is measured by Harrell’s C-index:

$$C = \frac{\sum_{i:\delta_i=1} \sum_{j:T_j>T_i} \mathbf{1}(\hat{r}_i > \hat{r}_j)}{\sum_{i:\delta_i=1} \sum_{j:T_j>T_i} 1} \quad (2)$$

3. Results

Table 1 summarizes concordance indices across all methods.

Table 1: Concordance index (C-index) by method. Values are mean $\pm$ SD over 10 cross-validation folds.

Method	C-index	IBS
Cox PH	0.632 ± 0.018	0.184 ± 0.009
Random Survival Forest	0.681 ± 0.015	0.162 ± 0.008
Gradient Boosting	0.688 ± 0.014	0.158 ± 0.007
Deep Survival	0.671 ± 0.017	0.169 ± 0.009

4. Discussion

The results in Table 1 confirm that ensemble methods outperform Cox regression on this benchmark. The Cox model (1) remains valuable for its interpretability (?).

5. Conclusion

Machine learning methods provide meaningful improvements over Cox regression. Future work should assess calibration.

References

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